

AION 4-20 mA Adapter - Telemetry Resources

Telemetry Resources - Group 0x02

Key	ID	Type	Default	Min	Max	Unit	Description
val1	0	float	0.0	-100000.0	100000.0	mA	Channel 1 raw loop current
val2	1	float	0.0	-100000.0	100000.0	mA	Channel 2 raw loop current
alarm	2	integer	0	0	255	bitfield	Per-channel alarm bitmask
status	3	integer	0	0	255	bitfield	Packed per-channel status codes
id	4	integer	0	0	65535	raw	Copy of configured id; the LPWAN payload currently transmits the low byte
upper1	5	float	10.0	-100000.0	100000.0	engineering value	Copy of configured channel 1 upper bound
upper2	6	float	10.0	-100000.0	100000.0	engineering value	Copy of configured channel 2 upper bound
temperature	1	integer	0	-100	100	degC	Internal temperature
lower1	7	float	0.0	-100000.0	100000.0	engineering value	Copy of configured channel 1 lower bound
lower2	8	float	0.0	-100000.0	100000.0	engineering value	Copy of configured channel 2 lower bound

Alarm bitmask in alarm

Bit	Mask	Meaning
0	0x01	Channel 1 scaled value is in alarm
1	0x02	Channel 2 scaled value is in alarm
2..7	0xFC	Reserved / currently unused

Status bitmask in status

Bits	Meaning
0..2	Channel 1 status code
3..5	Channel 2 status code
6..7	Reserved / currently unused

Channel status codes

Value	Name	Meaning
0	ok	Valid loop current
1	not_connected	Channel current below connection threshold; unused channels can be left disconnected
2	faulty	Under-range / sensor fault
3	overcurrent	Over-range / short condition
4	hw_fault	ADC read failed

Current classification in firmware

Raw current	Status
< 0.5 mA	not_connected
0.5..3.8 mA	faulty
> 3.8 mA and < 20.5 mA	ok
>= 20.5 mA	overcurrent